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Inventor(s)

Fakhar; Shaheen et al.

SYSTEMS AND METHODS FOR FAULT TOLERANT ENERGY MANAGEMENT SYSTEMS CONFIGURED TO MANAGE HETEROGENEOUS POWER PLANTS

Abstract

An energy management system and methods of using the same are disclosed. The energy management system is configured to deploy a first instantiation of a first hardware driver at a first electronic control device collocated with a first power plant apparatus, wherein the first hardware driver is configured to communicate with and enable control of the first power plant apparatus. The energy management system is configured to detect that the first instantiation of the first hardware driver at the first electronic control device has failed, and at least partly in response to detecting that the first instantiation of the first hardware driver at the first electronic control device has failed, utilize a second instantiation of the first hardware driver, stored in memory remote from the first electronic control device, to enable control of the first power plant apparatus.

Inventors: Fakhar; Shaheen (Herndon, VA), Toth; Robert Jordan (Leesburg, VA)

Applicant: Energy Vault, Inc. (Westlake Village, CA)

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

[0001] Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57.

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BACKGROUND OF THE INVENTION

Field of the Invention

[0003] The present disclosure generally relates to energy management systems.

Description of the Related Art

[0004] Energy management systems (EMSs) play a vital role in managing power plants. Energy management systems are platforms designed to monitor, control, and optimize energy usage, such as within power plants and utility grids.

[0005] However, conventional energy management systems often lack comprehensive monitoring, analysis, and control capabilities, and may be unable to manage heterogeneous power sources at the same plant. Further, many conventional energy management systems lack adequate security, thereby risking the secure and reliable operation of power plants. Yet further, conventional energy management systems rely on programmable logic controllers (PLCs) to provide control and automation capabilities for monitoring and managing energy-related processes. The use of PLCs often necessitates that EMS software be adapted to operation with a given PLC model.

[0006] Yet further, conventionally there are no adequate methods of ensuring a power plant site design will operate as expected when actually implemented. In addition, conventionally, site equipment drivers have to be constantly rewritten in order to accommodate new versions as well as old versions of power plant site equipment, such as programmable logic controllers.

[0007] These deficiencies may result in unreliable, expensive to maintain power plants and inefficient energy generation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Embodiments will now be described with reference to the drawings summarized below. These drawings and the associated description are provided to illustrate example aspects of the disclosure, and not to limit the scope of the invention.

[0009] FIG. 1 illustrates an example architecture.

[0010] FIG. 2 illustrates an example software architecture.

[0011] FIG. 3 illustrates an example power plant and EMS configuration.

[0012] FIG. 4 illustrates an example process and architecture.

[0013] FIG. 5 illustrates an example use of a power plant site digital twin.

[0014] FIG. 6 illustrates an example emulation-based test process and software deployment.

DETAILED DESCRIPTION

[0015] An aspect of the present disclosure relates to systems and methods for energy management.

[0016] As similarly discussed elsewhere herein, conventional energy management systems often lack comprehensive monitoring, analysis, and control capabilities, and may be unable to manage heterogeneous power sources at the same plant. Further, many conventional energy management systems lack adequate security, thereby risking the secure and reliable operation of power plants. Additionally, many conventional energy management systems lack adequate failure recovery mechanisms. Yet further, conventional energy management systems generally rely on programmable logic controllers (PLCs) to provide control and automation capabilities for monitoring and managing energy-related processes. The use of PLCs often necessitates that EMS software be adapted to operation with a given PLC model. Yet further, conventionally there are no adequate methods of ensuring a power plant site design will operate as expected when actually implemented. In addition, conventionally, site equipment drivers have to be constantly rewritten in order to accommodate new versions and well as old versions of power plant site equipment, such as programmable logic controllers.

[0017] The foregoing deficiencies may result in inefficient energy generation, unreliable operations, and inflexible infrastructure.

[0018] The disclosed systems and methods address these technical challenges. For example, disclosed herein is a hardware agnostic EMS which can be configured to manage heterogeneous power sources (e.g., gravity, solar, hydro, hydrogen, wind, gas etc.). In addition, the disclosed systems and methods provide enhanced security via the utilization of encrypted communications with distributed power plant controllers. The disclosed EMS comprises a highly scalable, technology-agnostic intelligent software solutions suite that provides real-time remote monitoring, operational control, and/or artificial intelligence-enabled optimized dispatch across an array of energy storage and generation assets. The disclosed hardware agnostic EMS is configured to coordinate equipment controls such as batteries, inverters, generators, and/or other apparatus to provide a cohesive operational view of a power plant while facilitating continuous data collection and analysis.

[0019] Further, the disclosed systems and methods provide enhanced failure recovery mechanisms, further enhancing reliability. For example, optionally the disclosed EMS system may be both locally hosted on a computer system at a power plant (which may be referred to as a site controller) and hosted on an offsite cloud-based system (which may be referred to as an EMS cloud).

Optionally, to further enhance reliability, in the event of a system failure of the local computer system hosting the EMS, the operation may failback to the cloud-based EMS system. The power plant controllers may thus be optionally controlled via the local EMS system or the cloud-based EMS system. The disclosed EMS may optionally utilize servers as controllers to be used in place of PLCs, which enables the virtualization of the monitoring and control functions needed to operate power generation infrastructure, thereby further enabling the EMS to be hardware agnostic.

[0020] Thus, the disclosed methods and systems overcome many technical challenges in managing energy systems.

[0021] Further, an aspect of the present disclosure relates to enabling a user to build a virtual power plant and to test the EMS against the virtual power plant to ensure the EMS software and the power plant design perform as expected or desired.

[0022] The sited-based EMS and/or the EMS cloud (which may at times be collectively referred to as the EMS) may communicate with various sensors, meters, and monitoring devices to collect real-time energy consumption, generation, and environmental data. For example, the EMS may optionally communicate with server-based controllers which provide enhanced programmability,

standardized hardware platform, and improved flexibility as compared to the conventional use of PLCs (although the EMS may be utilized with PLCs). The server-based controllers and/or PLCs may optionally be collocated (e.g., in very close proximity with the power plant equipment) with the equipment or devices being controlled by the controllers and/or PLCs.

[0023] The EMS may include a data analytics module that may be utilized to process and analyze data acquired (e.g., via controllers) via the various sensors, meters, and monitoring devices. As described herein the data analytics module may identify usage and operational patterns and trends, detect anomalies in energy usage, and/or detect component failures enabling the system to generate insights and recommendations for energy optimization and/or enhance the operation of a power facility.

[0024] The EMS may include a decision support module and a control module. For example, a decision support module may utilize the analysis to generate instructions and/or recommendations for managing electricity generation, load balancing, the use of backup equipment, and/or peak demand management. A control module may be utilized to implement the instructions and recommendations generated by the decision support module. For example, the control module may interface with energy-generation systems (e.g., gas, solar, wind, hydro, natural gas, coal, and/or other energy-generation systems), energy storage systems (e.g., gravity and kinetic energy-based energy storage systems, batteries, compressed air, and/or other energy storage systems), and/or grid infrastructure to regulate energy flows, switch between energy sources, and dynamically adjust energy generation patterns.

[0025] Optionally, the EMS is configured to provide user interfaces, such as graphical user interfaces for display on user terminals, that enable users to monitor energy consumption, monitor operational parameters of energy-related systems discussed herein, and adjust system settings. Optionally, user interfaces may be configured to provide real-time visualizations, reports, and alerts, enabling users to make informed decisions and actively participate in energy management of energy sources.

[0026] Certain aspects will now be described with reference to the figures.

[0027] Referring to FIG. 1, an example, EMS environment is illustrated. A cloud-based EMS computer system **102** is connected to a network **104** (which may be the Internet, or other public or private wide area network and/or local network). The cloud-based EMS computer system **102** may comprise a hosted computing environment that includes a collection of physical computing resources that may be remotely accessible, located at different facilities, and may be rapidly provisioned as needed (sometimes referred to as a “cloud” computing environment). Certain data described herein may optionally be stored using a data store that may comprise a hosted storage environment that includes a collection of physical data storage devices that may be remotely accessible and may be rapidly provisioned as needed (sometimes referred to as “cloud” storage). The cloud-based EMS computer system **102** may communicate via the network **104** with one or more site-based EMS systems **106** comprising one or more computer systems hosting EMS software, which may be situated locally at an energy production site.

[0028] The site-based EMS system **106** may comprise one or a plurality of servers, that may be run in one or more clusters, where a given server may optionally comprise a microserver to reduce the associated footprint and cost. A server cluster may be utilized that offers improved performance, scalability, availability, and reliability as compared to a conventional server. By distributing the workload across multiple servers, if one server fails, another server within the cluster may assume the workload seamlessly. This redundancy reduces downtime and ensures uninterrupted access to the EMS application.

[0029] A server cluster may utilize load balancing techniques to distribute incoming requests or traffic across multiple servers in the cluster. Load balancers within the cluster may intelligently allocate requests to servers based on factors such as server capacity, current load, and/or geographic proximity. This helps distribute the server workload evenly and prevents or reduces the chance that

a given server will be overloaded with tasks, improving EMS performance and responsiveness. Further, advantageously, server clusters provide scalability by enabling additional servers to be added to the cluster as demand increases. Such horizontal scaling enables the EMS to handle increasing workloads. As the cluster size grows, the overall processing power and capacity of the EMS increase, ensuring adequate resources to meet growing demands. Yet further, as similarly discussed above, server clusters provide fault tolerance by utilizing redundant components and failover mechanisms. If a server within the cluster experiences a hardware or software failure, the workload may be automatically shifted to other available servers.

[0030] The EMS software may comprise small lightweight pieces of code that run on virtualization, each in its own container, thus enabling code to be run on different hardware using containerization and virtual machines. Containerization packages applications and their dependencies into self-contained units called containers. The site-based EMS system **106** may include interface modules (e.g., drivers, proxies, clients, servers, and/or the like) that are used to communicate with various pieces of equipment located at the power plant site. The interface modules may be utilized to ensure that site equipment is operated within limits and provide data to the EMS **106** (and optionally EMS **102**). As will be discussed further, different versions of equipment may be associated with different interface module versions. Virtualization may be utilized, wherein in a virtual machine, virtual interface modules are used to emulate hardware devices, such as network and/or device interfaces.

[0031] The site-based EMS **106** (and optionally, the EMS cloud **102**) may further comprise a supervisor module, that controls the operations of the site. The supervisor module may be hosted on a single server cluster node or the supervisor function can be distributed across multiple cluster server nodes or across a site

[0032] A given site-based EMS system **106** may be server based, and may comprise one or more backup systems (e.g., one or more backup servers), some or all of which may optionally be incorporated into the same housing or racks as the primary EMS system. The EMS backup systems may be configured to ensure the availability and reliability of significant or critical energy management functions in the event of a primary system failure. The backup EMS systems may utilize redundancy and failover mechanisms to minimize downtime and maintain the continuous operation of energy management processes.

[0033] For example, the site-located backup systems may incorporate redundant hardware components such as servers, controllers, and/or network devices. Optionally, the redundant components are configured in parallel to the primary EMS system, continuously mirroring some or all of the data and functionality of the primary EMS system. For example, data (e.g., critical data and/or other data) may be synchronized between the primary and backup systems in real-time or near-real-time to replicate data from the primary system to the backup system. This ensures that the backup system has up-to-date information and can seamlessly take over in case of a primary system failure. Optionally, data may also be synchronized between the site-located EMS **106** (and/or the server-based controllers **108-1**, **108-2**, **108-3**, . . . **108-N** discussed herein) and the cloud-based EMS **102**.

[0034] Thus, advantageously, If the primary site-located EMS **106** fails, the backup system seamlessly takes over, ensuring uninterrupted operation. Optionally, if both the primary and backup EMS systems **106** fail, the cloud-based EMS **102** may seamlessly take over. Optionally, uninterruptible power supplies may be utilized to ensure continuous power availability to the site-located backup systems in the event of short-term power outages or voltage fluctuations.

[0035] Failover from the primary system, in the event of a primary system failure, transitions control and operation from the primary system to the backup system. Failover mechanisms may include automatic switches to redirect data traffic, activate backup servers, and/or reconfigure network connections.

[0036] Advantageously, optionally, both the site-based EMS system **106** and the cloud-based EMS

system **102** may communicate with one or more server-based controllers **108-1, 108-2, 108-3, . . . 108-N** (e.g., to provide and/or receive instructions and/or data, such as data described herein). For example, the server-based controllers **108-1, 108-2, 108-3, . . . 108-N** may be utilized by the cloud based EMS **102** and/or the site-based EMS **106** to perform data acquisition with respect to the loading, currents, temperature, pressure, battery utilization, and/or other parameters of power-generating equipment. The rate of data acquisition may be programmed, where different types of data may be collected at different rates. For example, power generation system data may be read every 50 ms, while power storage system data may be read every 1 second. By way of further example, the server-based controllers **108-1, 108-2, 108-3, . . . 108-N** may use the acquired data to monitor and control energy systems and equipment.

[0037] The server-based controllers **108-1, 108-2, 108-3, . . . 108-N** may be utilized to perform fault detection and diagnostics: For example, the server-based controllers **108-1, 108-2, 108-3, . . . 108-N** may be utilized in identifying faults or abnormalities in energy systems. By way of illustration, server-based controllers **108-1, 108-2, 108-3, . . . 108-N** may detect deviations from normal operating conditions, such as sudden energy spikes, equipment failures, or irregularities in power quality. When anomalies are detected, the server-based controllers may be configured to trigger alerts to the EMS, log data for analysis via the EMS, and be utilized to perform automated corrective actions. The server-based controllers **108-1, 108-2, 108-3, . . . 108-N** may be utilized to implement energy-saving strategies by adjusting equipment operation and/or optimizing scheduling. Optionally, if a given server-based controller has a failure, some or all of the functionality of the server-based controller may be provided by the EMS system **106**, the EMS system **102**, or another server-based controller in the fallback configuration.

[0038] The site EMS **106** may optionally communicate with the server-based controllers **108-1, 108-2, 108-3** via a conventional Modbus link. Modbus is a serial communication protocol that conventionally does not provide for secure, encrypted communication, risking the operational safety and reliability of the site. Optionally, to enhance communication security, data and/or instructions communicated between the site EMS **106** (or EMS cloud **102**) may be encrypted. By way of example, the Apache Kafka protocol may be adapted for use for communication between the site EMS **106** and the controllers. The Apache Kafka protocol may provide security features such as authentication and encryption, and may be configured to use mechanisms such as SSL/TLS for encryption and use authentication protocols such as SASL (Simple Authentication and Security Layer) to secure client-broker communication. Such a communication protocol enables reliable and scalable messaging within a distributed streaming platform. Further, such a communication protocol provides a framework for clients to interact with protocol brokers, enabling efficient data transfer. The protocol may be configured to handle high-throughput real-time data streams and support fault-tolerance and horizontal scalability.

[0039] The server-based controllers **108-1, 108-2, 108-3, . . . 108-N** may be associated with (and co-located with and/or physically adjacent to) respective different types of power sources, such as energy generation and/or storage systems **110-1, 110-2, 110-3, . . . 110-N**, which may be located at different areas/plots of the power generated site. The energy generation and/or storage systems **110-1, 110-2, 110-3, . . . 110-N** may comprise photovoltaic cell-based energy sources, gas power generator systems, hydrogen-based power generation systems, thermal power generation systems, coal-based power generation systems, gravity and kinetic energy-based energy storage systems, battery-based energy storage systems, compressed air storage systems, and/or other energy storage systems. Optionally, if a given controller located at one location fails, a controller at another site may take over some or all of the functionality of the failed controller.

[0040] The power plant site may be operated in a distributed manner. Advantageously, the EMS cloud **102** and/or site EMS **106** can run different operational programs for different site power sources, enabling different site power sources to produce different amounts of power.

[0041] Although the EMS cloud **102** and the site EMS **106** may optionally be capable of

performing substantially the same functions (e.g., in the event of a failure of the site EMS **106**), in normal operation the EMS cloud **102** and the site EMS **106** may optionally be configured to perform different tasks.

[0042] For example, the site EMS **106** may be configured to read and monitor the statuses of field equipment, control and optimize operation of field equipment, store historical data in a local historian database, perform diagnostics and protections for functions such as safety and system availability, provide a browser-accessible graphical user interface (GUI) via which users can view historical and current equipment, view the site EMS **106** statuses and alarms, manage configurations, and send commands to site equipment and devices. Optionally, the site EMS **106** may operate independently of the EMS cloud **102**.

[0043] By way of further example, the EMS cloud **102** may be configured to securely communicate (e.g., via an encrypted communication channel with secure authentication) with the site EMS **106** over the network **104** for synchronization, retrieve data stored at the power supply site for long-term storage in the cloud, track performance metrics and analytics (e.g., for a specified time period or over the lifetime of the power plant), provide software updates to the site EMS **106**, and/or provide a graphical user interface via which users can operate one or more power plant sites that are integrated with the EMS cloud **102**.

[0044] Optionally, users may be enabled to login to the site EMS **106** while connected directly to the site EMS **106**, or remotely through the EMS cloud **102** utilizing a secure portal that supports multi-factor authentication. Optionally, the user interface may communicate with the EMS cloud **102** and/or the site EMS **106** utilizing a secure web service to better ensure that only authorized users can command and control the power plant site.

[0045] Optionally, a user interface (displayed on a user terminal) may be hosted by the EMS cloud **102** that may be populated with historical data through a secure connection to the instance running on the cloud. Optionally, the user interface may provide real-time updates from the EMS and render alerts to faults, warnings, and/or changes in status that are relevant to the operation of the power plant site. Optionally, a user interface may be provided via the site EMS **106** or the EMS cloud **102** that enables a user to change the mode of the EMS and place site hardware in-service or out-of-service.

[0046] Referring now to FIG. 2, an example EMS software architecture is illustrated. As similarly discussed elsewhere herein, the EMS software architecture is configured to be highly scalable, available, and adaptable to work with different energy storage systems, including future energy storage systems, and a hybrid combination of different types of energy storage systems.

[0047] The site EMS system **202** hosts EMS software which may be executed on redundant hardware enabling the system to operate even in the event of a hardware failure while maintaining the speed needed to support real-time adjustments of power (P) and reactive power (Q) (the power that flows back from a destination toward the grid) to respond to changing grid frequency and voltage. The EMS network is optionally configured to be secure with the utilization of one or more firewalls **206** and **210** (e.g., configured to block malware and application-layer attacks and optionally with an integrated intrusion prevention system) and secured network switches (e.g., with unused ports and services disabled/made unavailable, with the switches in a locked area that is inaccessible to unauthorized personnel, where the switches are password protected, and/or the like). By way of illustration, the firewall may protect the system and only allow traffic that is specific to the remote operation of the site through the firewall. The EMS system **202** may comprise control block hardware that may be air-cooled and optionally configured as a standard indoor 19" rack or in an outdoor-rated enclosure.

[0048] The EMS system **202** may communicate with an operator control system **212** via the firewall **210** (although optionally, the operator control system **212** may be connected to a network switch directly, avoiding having to have a separate firewall configuration). By way of illustration, customer-specified routes to include communications to the RTU may be configured through the

firewall. Optionally, the RTU or customer programmed supervisory control and data acquisition (SCADA) and real time automation controller (RTAC) can be directly connected to the network switch.

[0049] For example, the operator control system **212** may comprise a SCADA system including a SCADA Data Gateway (SDG) that provides OPC server capabilities (that converts the hardware communicated data, such as from a PLC, into OPC protocol, to enable other software, such as SCADA, to access the data for the secure and reliable exchange of data in the industrial automation space) to a variety of SCADA protocols such as DNP3, Modbus, IEC 60870, and/or IEC 61850. The operator control system **212** may further comprise a remote terminal unit (RTU) that monitors the field digital and analog parameters and transmits data to the SCADA system (e.g., the SCADA master station). In addition, the EMS software **202** may communicate user interfaces and data and instructions to and/or from a user terminal (e.g., as webpages accessible via a browser hosted on the user terminal). For example, the user interfaces, instructions received from a user via the user interfaces, and/or data populating the user interfaces may include the example user interfaces, user instructions and data described elsewhere herein. The SCADA system and/or user terminal may be associated with and accessible by a power utility operator and/or an independent power producer to monitor site operation and to provide instructions to the EMS.

[0050] The system can be set to follow RTU commands or manual commands from the user. While the system is charging and discharging, it may balance the SOC (state of charge) continuously across some or all battery banks. The system can also take battery banks out of operation or place battery banks into operation.

[0051] In addition, the EMS system **202** may communicate with power plant site equipment **204**, where the EMS system **202** may transmit commands to the site equipment **204** and may receive status and/or other data from the site equipment **204**. For example, the site equipment **204** may include battery energy storage systems (BESSs) which may be used to compensate for power intermittency (e.g., as a result of equipment failure, or other unavailability of power, such as during the night for solar power), power conditioning systems (PCSs), auxiliary meters, power breakers, Balance of Plant (BoP) power plant components, safety systems, and/or the like. By way of example, optionally the EMS system **202** may communicate with power plant site equipment **204** via communication protocols discussed herein (e.g., Apache Kafka, MODBUS, etc.). Optionally, battery blocks are connected to a network switch through a connected ring network. Optionally, groups of battery blocks may be connected through a single fiber loop.

[0052] The site EMS **202** may communicate with the EMS cloud **208** remote monitoring services via the firewall **206**. Optionally, to enhance security, only routes that have been explicitly opened are available for communication with the EMS cloud remote monitoring services **208**. The EMS cloud remote monitoring services **208** software may provide functionality such as described elsewhere herein. For example, the EMS **202** may pass data to the cloud-based EMS remote monitoring services **208**, and the remote monitoring services **208** may provide insights into the data remotely (e.g., to the power plant operator/owner). The collected data may be utilized in protecting the power plant site equipment and in ensuring the power plant equipment operates within specified parameters while enhancing the efficiency and safety of the systems overall. Advantageously, the EMS remote monitoring services **208** is configured to increase safety of site personnel as well as to optimize site performance.

[0053] The EMS cloud **208** may generate and provide user interfaces **214**, **216**, via a user terminal. The user interfaces **214**, **216** may be populated with analytics data and enable operation and maintenance (O&M) respectively.

[0054] For example, a user interface may graphically depict, via a single line drawing (SLD), a symbolic representation of an architecture of a power plant being managed via the EMS, and which may depict the paths for power flow (including real power in megawatts (MW) and reactive power in megavolt amps (MVR)) between entities of the system. By way of illustration, if a direct current

system is being represented, the line may include the supply and return paths, while in a three-phase system, the line may represent all three phases. The SLD may include symbolic representations of electrical elements such as circuit breakers, switchgear, transformers, surge arresters, capacitors, electrical bus bars, conductors, and the like.

[0055] For the AC power, the user interface may also depict in real time, the megawatts, the megavolt amps the megavolt amperes (MVA), and the power factor (PF) (the ratio of working power to apparent power) In addition, the user interface may report the current AC setpoints (target or desired operational parameters) and the most recent setpoint commands (e.g., the last setpoint command, the last 5 setpoint commands, or other specified number of previous setpoint commands). New setpoints may be entered prompting the EMS (e.g., the site EMS) to interact with devices and equipment (e.g., batteries, inverters, generators, and/or other apparatus) to modify operation (e.g., voltage, frequency, reactive power, power factor, ramp control, etc.) to achieve the new setpoint. The user interface may report site meter data, such as real power, reactive power, apparent power, power factor, grid frequency, three phase AC sinusoidal voltages (e.g., line-line voltages AB, BC, and CA). The AC status may be reported via the user interface, such as the AB, BC, and CA line-line voltages and the currents for the respective three phases. For DC power, the line-to neutral (L-N) voltages and current may be reported.

[0056] The user interface may also graphically and textually depict the nominal power plant energy capacity (e.g., a designed and installed capacity, and maximum producing capacity under specified conditions that is the full-load sustained output), in megawatt-hours (MWh) and/or kiloampere hours (kAh) and the real time percentage utilization thereof. Optionally, the current available energy capacity may be reported in real time. Optionally, the user interface may render graphs that depict, over a specified period of time (e.g., the last hour, the last 3 hours, the last 24 hours, etc.), the power plant real power and reactive power setpoint. Optionally, controls may be provided via which a user can add additional data points.

[0057] Optionally, the site-based EMS **106** may be configured with dynamically deployed, run-anywhere microservices, where microservices may be deployed and managed in a flexible and scalable manner, enabling the microservices to run on various platforms and environments. Such an approach provides scalability and portability in a distributed system architecture. For example, containerization may be used to enable run-anywhere microservices, where containers provide a lightweight and isolated runtime environment for the microservices, making them portable across different operating systems and cloud platforms.

[0058] An orchestration platform may be utilized to define the desired state of the microservices, handle automatic scaling, and distribute the workload across multiple containers or server nodes. A service discovery mechanism may be utilized to enable microservices to locate and connect to each other dynamically. A configuration management module may be utilized to configure certain microservice parameters, such as database credentials, API keys, and/or external service endpoints.

[0059] Optionally, a given site may be defined in software by a single configuration of site equipment and devices that specifies how they are electrically connected. Such specified configuration enables large scale deployment of digital twins (a digital representation of some or all of the site, and equipment and device therein). The digital twin may act like the power plant equipment from a software interface standpoint and where the digital twin may be indistinguishable to the EMS from the physical equipment with respect to the respective software interfaces (which may be identical). The digital twin enables a high fidelity model of the site to be deployed for simulation, integration, testing, monitoring, and/or maintenance. The site may be automatically deployed to a test framework enabling the site software to be tested in its final configuration before being deployed to the physical site. For example, based on the results of the automatic testing (e.g., the successful result) the configuration can be automatically deployed to a physical site immediately or in response to detecting a maintenance timeframe. Optionally, the configuration may be deployed manually.

[0060] For example, referring to FIG. 5, a digital twin **502** of a large battery storage site is illustrated. The digital twin **502** illustrates the connection between PLCs and batteries, between the battery banks and a circuit breaker, and between the circuit breaker and a load. The EMS software **504** communicates commands to the digital twin **502** and receives data, such as status data, from the digital twin **502** as if it were a physical, large battery storage site. The EMS software **504** may receive commands from a user via an operation and maintenance (O&M) user interface presented on a user terminal, which it may in turn relay the commands to the digital twin **502**. Similarly, the EMS **504** may populate the user interface with data (e.g., status data) received from the digital twin **502**. By way of further example, the EMS **504** may populate the user interface with time series data based on an analysis by the EMS software **504** of data received from the digital twin **502**.

[0061] Time series data analysis in energy management systems may provide a comprehensive understanding of energy consumption, load patterns, renewable energy integration, anomaly detection, and performance optimization. For example, time series data may be used to track and monitor energy consumption patterns over time. By collecting and analyzing energy consumption data at regular intervals, the EMS may identify consumption trends, peak demand periods, and energy usage patterns. This information may aid in optimizing energy usage and identifying opportunities for energy efficiency improvements. Time series data may be used to identify anomalies or abnormal patterns in energy consumption or system behavior. By comparing real-time data with historical data, the energy management system can detect deviations, spikes, or unusual trends that may indicate equipment malfunctions, energy wastage, or other operational issues. Time series data may also be used to evaluate and monitor the performance of energy assets, such as generators or transformers, over time. By analyzing trends and deviations in performance metrics, the EMS may identify inefficiencies, assess equipment health, and optimize energy asset utilization.

[0062] As similarly discussed elsewhere herein, dynamically deployed device interface modules may be utilized to enhance fault tolerance and flexible deployments. Optionally, dynamically deployed controllers are utilized to provide fault tolerance and functional improvements through their ability to balance workload, detect failures, recover from faults, provide service discovery and routing, support auto-scaling, and facilitate dynamic reconfiguration. Such features enhance the system's resilience, scalability, availability, and performance.

[0063] The dynamic deployment of device interface modules in an EMS overcomes technical challenges not conventionally addressed. In the industrial control world, and in particular power plants, conventionally many device interface modules and control devices are controlled by software that runs on dedicated equipment or simple programmable logic controllers. Disadvantageously, conventionally interface modules are thus tied to specific hardware in control systems. To the extent that failover capabilities are provided in such conventional approaches, typically those failovers are onto a specific backup device that runs on standby. This rigid approach may complicate deployment of the software and necessitate redundant equipment.

[0064] By contrast to conventional approaches, dynamically deployed EMS hardware control device interface modules may be used to control hardware devices that can be run in multiple systems. For example, dynamically deployed device interface modules may be co-located with the equipment they are to communicate with, be executed by an on-site computer system co-located with equipment, or be executed to run remotely in the cloud (e.g., the EMS cloud **102**). The ability to deploy EMS hardware control interface modules that can run on different systems and are not restricted to specific hardware provide several technical improvements over conventional approaches. For example, the need for redundant equipment (e.g., computers that communicate with power plant equipment such as inverters and battery storage systems) may be greatly reduced (although such redundant equipment may be used). By way of illustration, using the described technology, when a computer co-located with power plant equipment fails, the EMS may dynamically execute (on the EMS computer system) a virtual interface module that controls the

power plant equipment. Further, the virtual interface modules may be executed on multiple on-site computers and thus a cluster of computers may provide many combinations of redundancy to control individual items of site equipment and devices. Optionally, the computer selected to execute an interface module may be selected so as to reduce or minimize latencies between the computer and the equipment or device for which the interface module will be used.

[0065] Referring now to FIG. 3, an example storage site architecture comprising a system of batteries **302**, wind turbines **304** (having an associated interface module), and solar power generator **306** (having an associated interface module), is illustrated. In this example, battery banks are associated with respective PLCs (although some or all of the PLCs may be replaced or augmented with a server or other computer system performing PLC functions). As will be described, there may be multiple instantiations of a given device interface module (e.g., in the memory of one or more controller devices, in a site-based EMS, in an EMS cloud, and/or the like).

[0066] For example, a given PLC (or a server performing PLC functions) is associated with a respective interface module. The interface module may be selected by a control node of the site-based EMS system **308**, which may store different interface modules for different devices and equipment. For example, a site-based EMS system **308** supervisor module may determine which interface module to deploy based on the configuration (e.g., the version and/or manufacturer of the PLC) and runtime conditions. As similarly discussed elsewhere herein, if a interface module associated with a PLC fails (e.g. as a result of a bug or programming error, as a result of a software update which is incompatible with the PLC, as a result of a lack of system resources or PLC hardware failure, etc.), the site-based EMS system **308** may select and execute the interface module using EMS system **308** resources (e.g., processing bandwidth and memory). Optionally, for test purposes or in the event the site-based EMS system **308** has failed, and a failover to the EMS cloud **310** has taken place, the EMS cloud **310** may select and execute a PLC interface module (e.g., in the event of a failed PLC interface module).

[0067] Optionally, the EMS system **308** may be set by a user via a user interface to follow RTU commands or to carry out manual commands from the user. Optionally, while the battery system **302** is charging and discharging, the state of charge (SOC) may be balanced continuously across the various battery banks, where the SOC is the level of charge of an electric battery relative to its capacity. Optionally, the EMS system **308** can, automatically or under manual control, place battery banks in operation or take battery banks out of operation.

[0068] The disclosed software may be reusable, flexible software composed in one or more transportable languages (e.g., JAVA, PYTHON, etc.) that can run on small, embedded computers, enterprise servers, and/or cloud-based computer deployments.

[0069] In the event an item of software (e.g., an interface module) is needed, the EMS may detect the availability of the software and then launch backup instances either in hot standby or on-demand. For example, hot standby software may be used to ensure high availability and fault tolerance by maintaining a redundant backup that can seamlessly take over operations in case a primary system fails. The hot standby software may monitor the health and availability of the primary system (e.g., checking system metrics, network connectivity, application responsiveness, and/or other relevant indicators). If the primary system is detected as unavailable or experiencing issues, the hot standby software may trigger the failover process. During failover, the backup software may take over the workload and operations seamlessly, assuming the role of the primary system. This failover transition may optionally include redirecting network traffic, activating services, and ensuring data consistency.

[0070] Optionally, the EMS determines the best virtual interface module to deploy depending on the current conditions and use cases. For example, if a computer co-located with a power plant equipment or device asset is available to execute the virtual interface module, the drive may be executed on such computer. If the computer is determined to be unavailable, the interface module may be executed on the site EMS or by the EMS cloud. Thus, advantageously, dedicated or specific

backup computers are not required to ensure continuity of operations on the site. In addition, costs, energy consumption, and the hardware footprint of the site may be reduced.

[0071] Optionally, multiple computers may be distributed across the site. Thus, if there is a physical failure of a sector of the site (e.g. a failure of a given computer that controls site equipment or a disruption in auxiliary power), the control of the affected site equipment or devices may fail-over to another computer that may execute the appropriate interface module.

[0072] Thus, the disclosed architecture may increase the availability of the power plant site (e.g., approaching 100% availability) and provides enhanced fault tolerance as compared to an architecture that relies on dedicated standby PLCs or servers. Further, the disclosed architecture may provide enhanced flexibility with respect to the hardware deployed to the site. Still further, the disclosed techniques facilitate collocating interface modules with the hardware in cases where the hardware interfaces may be insecure, enabling the interface module to provide a secure interface into the equipment or device.

[0073] Further, the dynamic deployment controllers may be utilized to provide fault tolerance and functionality improvements. By way of example, a conventional power generator plant is run as a single operation. Conventionally, supervisory controllers run on a single server, sometimes with provisions for a hot backup. More recently, the concept of virtual power plants has been developed. A virtual power plant may couple multiple power plant assets as a single resource. The disclosed hardware and software architecture, and techniques may improve the operation of a virtual power plant by better facilitating the dynamic distribution of energy from multiple power generators in real time, such as behind the meter assets (e.g., distributed energy resources, such as solar panels and energy storage system, such as batteries).

[0074] Further, software controllers may be dynamically deployed in different environments. Using the disclosed hardware and software architecture, and techniques, a power generator may be dynamically coupled to other assets, such as behind the meter resources as part of a virtual power plant. Thus, rather than be limited to transmitting commands to nodes in the virtual power plant, the control software itself can also be distributed between power generator assets and may provide control of multiple assets in the network.

[0075] As discussed elsewhere herein, a server cluster may be utilized that offers improved performance, scalability, availability, and reliability as compared to a conventional server. The nodes may be deployed across a cluster and that can be further extended into cascading commands to individually deployed assets. For example, cascading commands may be utilized to propagate commands or actions from one node or server in the cluster to other nodes in a sequential or cascading manner. Cascading commands may be utilized to manage and coordinate the behavior of the cluster as a whole.

[0076] For example, a lead node may be configured to take over a given use case and delegate to follower nodes across multiple sites as needed. Such delegation enables the distribution of workload, improves fault tolerance, and enhances overall cluster performance. The lead node may determine the tasks that need to be performed within the cluster. Examples of tasks include processing incoming requests, executing computations, managing resources, taking equipment off line or placing equipment on line, or coordinating cluster operations. The use of a server cluster and cascading commands may be used in conjunction with the interface module framework described herein to enable the site to be dynamically controlled across different compute assets, including assets that may come online or go offline over time. A remote node may serve as a gateway for remote control ensuring authorized tasking is passed to the site lead node. The remote node may optionally also serve as a backup to the lead node and may also perform as a follower node. The use of a server cluster and lead and follower nodes may be scaled to massive virtual power plants with multiple layers of leads and followers.

[0077] A given cluster node may be located at the power plant it is being used to control aspects of, may be located at another power plant and used to control aspects of two or more power plants, or

may be cloud-based and used to control equipment at one or more power plants.

[0078] Referring to FIG. 4, an example process and architecture are illustrated. Although microcontrollers may be referred to in this example, other types of appropriately configured computing devices may be utilized. The microcontrollers may be configured for use in power generation facilities (e.g., specialized microcontrollers designed for use in power generation facilities). A given microcontroller may comprise an embedded system that incorporates a microprocessor (e.g., configured to handle complex calculations and control algorithms needed for monitoring and controlling power plant processes), memory, input/output interfaces, and other components (optionally on a single chip). The input/output interfaces may be configured to connect with sensors, actuators, and other power plant equipment. The input/output interfaces may include analog-to-digital converters (ADCs), digital-to-analog converters (DACs), serial communication ports, pulse-width modulation (PWM) outputs, and/or the like. A given microcontroller may support one or more communication protocols (e.g., Modbus, Profibus, Ethernet, and/or the like) to enable data exchange with other power plant systems (e.g., the EMS) and devices. A given microcontroller may be equipped with a real-time operating system (RTOS) or a real-time kernel to handle time-critical power plant tasks.

[0079] A given microcontroller may be used to monitor and control various processes, such as fuel supply, combustion control, turbine speed, and/or generator voltage regulation. Using the input signals from sensors, the microcontroller may ensure that the power plant operates within safe and efficient limits. The sensor data may include information related to temperature, pressure, flow rates, voltage, current, and other relevant parameters. Such data may be utilized to monitor the plant's performance, to detect abnormalities, and perform operation optimization to enhance plant efficiency.

[0080] A user **400** may task the system to perform operations via the EMS cloud **408** via commands submitted via a user interface or file. An EMS cloud **408** remote control node, configured to perform remote control and hot standby functions, may pass the tasks to an EMS site microcontroller lead. The EMS site microcontroller lead may then perform real time distribution of tasks to EMS microcontroller followers. Site microcontroller nodes (which may be part of site-located EMS server cluster), comprising the EMS site microcontroller lead and EMS site microcontroller followers, may be distributed across geographic regions (e.g., parcel 1 **402**, parcel 3 **406**) or be collocated. For example, the battery system of parcel 2 **404** may be controlled directly by the EMS site microcontroller lead located at parcel 1 **402**. The collection of controllers may run in a cluster and may support failover and may be used to distribute the processing load amongst the controllers. The lead node receives tasks for the site and appropriately delegates tasks to the appropriate follower nodes to ensure the corresponding operations are performed. Certain nodes may be in hot standby (such as the EMS hot standby node), ready to seamlessly assume the tasks of another node in the event of a node outage.

[0081] Optionally, multiple user signals may pass tasking to the EMS. For example, user signals may define multiple ways for power to be commanded, serving multiple customers. A given system or parcel may be metered and the metering from the different systems or parcels may be combined to form virtual meters to ensure power is being distributed as needed.

[0082] A technical problem introduced by using offsite software instances, such as interface modules, to control site assets (e.g., equipment and devices) in the event of an on-site failure, is that latencies may be introduced in communicating commands to power plant assets and in receiving data from assets. Optionally, the assets may be controlled at a slightly slower rate to accommodate such latencies, however, taking such assets offline may be avoided even when site computers (e.g., hosting the site-based EMS) fail and are not operational.

[0083] Certain concepts, similar to those discussed elsewhere herein with respect to the automatic deployment of software assets may be applied to a test and deployment infrastructure.

[0084] A power plant site may be defined by configuration. The configuration may be applied to

determine how the site is deployed as well as the creation of a test framework. Conventionally, energy storage and industrial control systems are large and so tend to be difficult to test. In addition, with respect to testing battery storage systems, batteries are difficult to store prior to deployment, and experience degradation during test. Further, appropriate test equipment is expensive and may even be unavailable due to limited supply and heavy market demands.

[0085] In order to address the lack of equipment to test prior to deployment, a robust test framework is needed. Such a test framework will now be described. An emulation framework is provided that emulates some or all of the equipment onsite at the power plant. The test framework enables the modeling of equipment and devices that are electrically connected and the modeling of when a given device is operating. In addition, the behavior of the sensing aspects of the site, including meters, may be modeled. For example, the sensing of power/reactive power, current, voltage, temperature may be modeled.

[0086] Further, deployment challenges presented by the use of various types of equipment from various equipment vendors may be addressed via a vendor interface module library. For example, multiple versions of the interface module library may be developed with a software emulator. Thus, rather than attempting to have a single interface module work with older and newer versions of a PLC, each version of the PLC may have its own interface module. Thus, an older version of the PLC may be utilized with an older version of the interface module compatible with the older version of the PLC, and a newer version of the PLC may be utilized with a newer version of the interface module compatible with the newer version of the PLC. This approach may enhance reliability, while reducing the need to constantly modify interface module code.

[0087] A site configuration registry may be generated and stored in memory that defines some or all of the following: [0088] hardware on the site [0089] the electrical connection between site equipment and assets [0090] the applicable use cases of the software [0091] limits for the equipment and site operations [0092] what interface module version to deploy

[0093] A site configurator software program may access the site registry from memory and deploy the site configurator to both test and operational sites. A DevOps pipeline may be defined that enables the site configuration to be automatically used to test a virtual site and then automatically deploy to the site remotely. A DevOps pipeline, also known as a CI/CD pipeline (Continuous Integration/Continuous Deployment pipeline), may comprise a set of automated processes and tools that facilitate the development, testing, and deployment of software applications.

[0094] Thus, because a site may optionally be defined in single configuration file, the software may be deployed to site equipment or test computers (in the cloud or onsite) and the software may be run against either real physical site equipment, emulated equipment, or a mix of real physical site equipment and emulated equipment (where physical equipment that is not yet onsite or that is otherwise unavailable for testing may be emulated). Advantageously, optionally, the real equipment and emulated equipment may be indistinguishable to the EMS software, enabling the emulated equipment to be exercised as if it's actual operation.

[0095] The emulation framework enables the scaling of very large physical sites in scalable cloud servers, enabling multi-GWh sites to be simulated in virtual operations prior to deployment to the site. This enables the site capabilities to be demonstrated to end users (e.g., power plant site operators) well in advance of physical deployment, as well as providing the opportunity to resolve software issues and deficits prior to deployment. Further, and particularly with respect to large power plant sites, emulation of the site modeled with respective equipment and devices electrically connected and deployed to the cloud, enables a higher fidelity virtual digital twin deployment which better provides a fuller demonstration of capabilities of the EMS without needing any of the physical site equipment and devices.

[0096] Additional orchestration (e.g., managing and coordinating the deployment and execution of tests in a test environment) enables the efficient and automated management of various testing components, such as test cases, test data, test environments, and test execution and the deployment

of the software in a test configuration. The test configuration can be tested with test commands and use cases passed to the EMS via the public endpoints and/or via the emulation framework. Based on the test results (e.g., the success or failure of the automated tests, which may be determined by a comparison of expected or desired power plant site behavior with the simulated power plant site behavior), a decision may be made as to whether the software is ready to be deployed to the hardware at site.

[0097] By way of illustration, if aspects (e.g., operating parameter values) of the simulated behavior differ by more than corresponding threshold amounts from expected behavior (e.g., expected operating parameter values), a failure indication may be generated, and the software may not be deployed. If, instead, if aspects of the simulated behavior do not differ by more than corresponding threshold amounts from expected behavior, a pass indication may be generated, and the software may be deployed. Such aspects may relate to the temperature of various power plant apparatus, the state of charge of a battery energy system, real power, reactive power, apparent power, power factor, three phase AC sinusoidal voltages, and/or other aspects, such as those disclosed herein.

[0098] For example, if the software successfully passes the automated testing, the software can then be deployed to the hardware at the power plant site (e.g., automatically, using scripted deployment, or manually). As similarly discussed elsewhere herein, the software may be substantially immediately deployed to the site upon passing testing, or the software may be downloaded to memory of the site-located computer hosting the EMS, and may be installed and deployed at the next available scheduled maintenance time frame or other specified time.

[0099] Referring now to FIG. 6, a site configurator **606** accesses a site configuration registry (e.g., that defines hardware on the site, the electrical connections between site equipment and assets, the applicable use cases, limits for the equipment and site operations, what interface module version to deploy, and/or the like) and creates emulators **604** for the corresponding hardware on the EMS cloud **602**, where the EMS cloud **602** can interact with the emulation of the power plant site hardware. The site hardware (which may include software) that may be emulated may include battery energy storage systems (BESSs), power conditioning systems (PCSs), auxiliary meters, power breakers, Balance of Plant (BoP) power plant components, safety systems, communication buses and protocols (e.g., Apache Kafka, MODBUS, etc.), and/or the like. The emulation enables large-scale testing of the site defined by the site configuration registry (which may correspond to both the emulated equipment and the actual, physical equipment).

[0100] A system automated test module **606** may inject user cases (e.g., loads, etc.) into EMS endpoints, the emulated equipment may respond to the use cases, and the automated test module **610** may monitor the emulated equipment to determine whether they are operating as expected or desired. Example endpoints may include data acquisition endpoints responsible for collecting data from various sources within the energy system, control system interfaces endpoints that enable the EMS **602** to interact with control devices or equipment, energy storage endpoints that enable monitoring and control of energy storage systems (e.g., charging, discharging, state-of-charge information, etc.), renewable energy interface endpoints, demand response endpoints that enable the interaction between the EMS **602** and external entities such as utility companies or demand response aggregators (e.g., where the endpoints receive signals or requests for load reduction or load shifting during peak demand periods to enable the EMS **602** to curtail loads, adjust energy consumption, or participate in demand response programs), communication gateway endpoints and/or other endpoints.

[0101] Once the emulated site adequately passes the tests, an automated deployment process **612** may be so informed, and the automated deployment process **612** may in response automatically deploy the software to the physical power plant site **614** (e.g., to the onsite EMS system **616**) as authorized (e.g., immediately, at the next maintenance window, other schedule time). Optionally, the software deployment may be performed manually, in response to a user or present disclosure

relates to an energy management system whose functionality can be dynamically shifted from a local computer system to a cloud system.

[0102] Thus, an aspect of the present disclosure relates to a site energy management system that can dynamically be adapted to different energy sources, including battery and hydrogen energy sources, and that can switch from one energy source to another in the event of a failure. Another aspect of the present disclosure relates to methods for testing a site energy management system via a digital model of the site energy management system. Yet another aspect of the present disclosure relates to a site energy management system whose functionality can be dynamically shifted from a local computer system to a cloud system. Still another aspect of the present disclosure relates to various failover mechanisms that enhance site availability. Another aspect of the present disclosure relates to the use of interface module libraries to reduce the need to constantly rewrite interface module code to accommodate multiple versions of a device.

[0103] An aspect of the present disclosure relates to an example energy management system and methods of using the same. The energy management system is configured to deploy a first instantiation of a first hardware driver at a first electronic control device collocated with a first power plant apparatus, wherein the first hardware driver is configured to communicate with and enable control of the first power plant apparatus. The energy management system is configured to detect that the first instantiation of the first hardware driver at the first electronic control device has failed, and at least partly in response to detecting that the first instantiation of the first hardware driver at the first electronic control device has failed, utilize a second instantiation of the first hardware driver, stored in memory remote from the first electronic control device, to enable control of the first power plant apparatus.

[0104] Another aspect of the present disclosure relates to an energy management system, comprising: a network interface; at least one processing device operable to: deploy a first instantiation of a first hardware interface module at a first electronic control device collocated with a first power plant apparatus, wherein the first hardware interface module is configured to communicate with and enable control of the first power plant apparatus; detect that the first instantiation of the first hardware interface module at the first electronic control device has failed; at least partly in response to detecting that the first instantiation of the first hardware interface module at the first electronic control device has failed, utilize a second instantiation of the first hardware interface module, stored in memory remote from the first electronic control device, to enable control of the first power plant apparatus via the network interface.

[0105] Optionally, the first power plant apparatus, the first electronic control device, and the memory remote from the first electronic control device, are located at a first power plant site, wherein the at least one processing device is operable to: enable a third instantiation of the first hardware interface module, hosted by a cloud-based computer system remote from the first power plant site, to enable control of the first power plant apparatus at least partly in response to a failure of the first instantiation of the first hardware interface module and a failure of the second instantiation of the first hardware interface module. Optionally, the memory remote from the first electronic control device comprises memory of a second electronic control device. Optionally, there are a plurality of instantiations of the first hardware interface module stored in memory of a plurality of electronic control devices, wherein the second instantiation of the first hardware interface module is selected to enable control of the first power plant apparatus based at least in part on its proximity to the first power plant apparatus. Optionally, the second instantiation of the first hardware interface module is maintained in hot standby or on-demand mode. Optionally, the first power plant apparatus comprises a battery or an inverter. Optionally, a plurality of versions of the first electronic control device are located at a first power plant site, wherein the at least one processing device is operable to: select from a plurality of versions of hardware interface modules stored in memory, the plurality of versions of hardware interface modules configured for respective versions of the first electronic control device, an appropriate version of the hardware device

interface module for a first version of the first electronic control device; and deploy the appropriate version of the hardware device interface module for the first version of the first electronic control device to the first version of the first electronic control device. Optionally, the first electronic control device comprises a programmable logic controller. Optionally, the first instantiation of the first hardware interface module failed as a result of a failure of the first electronic control device. Optionally, the second instantiation of the first hardware interface module is maintained in hot backup execution mode.

[0106] Another aspect of the present disclosure relates to an energy management system, comprising: a network interface; at least one processing device operable to: deploy a first instantiation of a first hardware interface module at a first electronic control device collocated with a first power plant apparatus, wherein the first hardware interface module is configured to communicate with and enable control of the first power plant apparatus; detect that the first instantiation of the first hardware interface module at the first electronic control device has failed; at least partly in response to detecting that the first instantiation of the first hardware interface module at the first electronic control device has failed, utilize a second instantiation of the first hardware interface module, stored in memory, to enable control of the first power plant apparatus via the network interface.

[0107] Optionally, the first power plant apparatus, the first electronic control device, and the memory, are located at a first power plant site, wherein the at least one processing device is operable to: enable a third instantiation of the first hardware interface module, hosted by a cloud-based computer system remote from the first power plant site, to enable control of the first power plant apparatus at least partly in response to a failure of the first instantiation of the first hardware interface module and a failure of the second instantiation of the first hardware interface module. Optionally, the memory comprises memory of a second electronic control device. Optionally, the memory comprises memory of a second electronic control device collocated with the first electronic device. Optionally, the memory comprises memory of a second electronic control device remote from the first electronic device. Optionally, there are a plurality of instantiations of the first hardware interface module stored in memory of a plurality of electronic control devices, wherein the second instantiation of the first hardware interface module is selected to enable control of the first power plant apparatus based at least in part on its proximity to the first power plant apparatus. Optionally, the second instantiation of the first hardware interface module is maintained in hot standby or on-demand mode. Optionally, the first power plant apparatus comprises a battery or an inverter. Optionally, a plurality of versions of the first electronic control device are located at a first power plant site, wherein the at least one processing device is operable to: select from a plurality of versions of hardware interface modules stored in memory, the plurality of versions of hardware interface modules configured for respective versions of the first electronic control device, an appropriate version of the hardware device interface module for a first version of the first electronic control device; and deploy the appropriate version of the hardware device interface module for the first version of the first electronic control device to the first version of the first electronic control device. Optionally, the first electronic control device comprises a programmable logic controller. Optionally, the first instantiation of the first hardware interface module failed as a result of a failure of the first electronic control device.

[0108] Another aspect of the present disclosure relates to systems and methods for testing a power plant emulation. Software for an energy management system is deployed to a computer system comprising computing devices. A model of a power plant site is deployed to a test framework. Test commands and/or use cases are injected, via the test framework to public energy management system end points. The test commands and/or use cases are used to emulate and test the model of at least the portion of the power plant site and to generate simulated operating parameter values. The generated simulated operating parameter values of the model are compared with expected values. A determination is made as to whether the model passed testing using the comparison. After

determining the model passed testing, software, corresponding to at least a portion of the model, is made available for deployment to operate a physical version of the power plant site.

[0109] Another aspect of the present disclosure relates to a computer-implemented method of testing a power plant emulation, the method comprising: deploying software of an energy management system to a computer system comprising one or more computing devices; deploying a model of at least a portion of a power plant site to a test framework; injecting, via the test framework, test commands and/or use cases to public endpoints of the energy management system; using the test commands and/or use cases to emulate and test the model of at least the portion of the power plant site and to generate simulated operating parameter values; comparing the generated simulated operating parameter values of the model with expected values; determining whether the model passed testing based at least in part on the comparison of the generated simulated operating parameter values of the model with the expected values; and at least partly in response to determining the model passed testing, automatically make software, corresponding to at least a portion of the model, available for deployment to operate a physical version of the power plant site.

[0110] Optionally, the model comprises a digital twin with respect at least a portion of physical equipment at the power plant, the digital twin having a software interfaces indistinguishable from software interfaces of corresponding physical equipment. Optionally, the public endpoints of the energy management system comprise data acquisition endpoints configured to collect data from sensors at the power plant site. Optionally, the model of at least the portion of a power plant site is specified via a site configuration registry that defines a power plant apparatus, electrical connections between the power plant apparatus, and apparatus limits. Optionally, the model of at least a portion of the power plant site is specified via a site configuration registry, the method further comprising, at least partly in response to determining the model passed testing, deploying the site configuration registry to the physical version of the power plant site. Optionally, the software, corresponding to at least a portion of the model, is deployed to operate the physical version of the power plant site during a maintenance window. Optionally, the software, corresponding to at least a portion of the model, is deployed to operate the physical version of the power plant site substantially immediately after being made available for deployment. Optionally, the model comprises models of battery energy storage system, a power conditioning system, a meter, and a power breaker.

[0111] Another aspect of the present disclosure relates to computer-implemented method of testing a power plant emulation, the method comprising: deploying software for an energy management system to a computer system comprising one or more computing devices; deploying a model of at least a portion of a power plant site to a test framework; injecting, via the test framework, test commands and/or use cases to public endpoints of the energy management system; using the test commands and/or use cases to emulate and test the model of at least the portion of the power plant site and to generate simulated operating parameter values; comparing the generated simulated operating parameter values of the model with expected values; determining whether the model passed testing based at least in part on the comparison of the generated simulated operating parameter values of the model with the expected values; and at least partly in response to determining the model passed testing, performing a first action.

[0112] Optionally, the model comprises a digital twin with respect at least a portion of physical equipment at the power plant, the digital twin having respective software interfaces indistinguishable from software interfaces of corresponding physical equipment. Optionally, the public endpoints of the energy management system comprise data acquisition endpoints configured to collect data from sensors at the power plant site. Optionally, the model of at least the portion of a power plant site is specified via a site configuration registry that defines a power plant apparatus, electrical connections between the power plant apparatus, and apparatus limits. Optionally, the model of at least a portion of the power plant site is specified via a site configuration registry, the method further comprising, at least partly in response to determining the model passed testing,

deploying the site configuration registry to a physical version of the power plant site. Optionally, software, corresponding to at least a portion of the model, is deployed to operate a physical version of the power plant site during a maintenance window. Optionally, software, corresponding to at least a portion of the model, is deployed to operate a physical version of the power plant site substantially immediately after being made available for deployment. Optionally, the model comprises models of battery energy storage system, a power conditioning system, a meter, and a power breaker.

[0113] Another aspect of the present disclosure relates to a non-transitory, tangible, computer readable memory that stores program instructions that when executed by a computer system comprising a computing device, cause the computer system to perform operations comprising: deploying software for an energy management to a computer system comprising one or more computing devices; deploying a model of at least a portion of a power plant site to a test framework; injecting, via the test framework, test commands and/or use cases to public endpoints of the energy management system; using the test commands and/or use cases to emulate and test the model of at least the portion of the power plant site and to generate simulated operating parameter values; comparing the generated simulated operating parameter values of the model with expected values; determining whether the model passed testing based at least in part on the comparison of the generated simulated operating parameter values of the model with the expected values; and at least partly in response to determining the model passed testing, performing a first action.

[0114] Optionally, the model comprises a digital twin with respect at least a portion of physical equipment at the power plant, the digital twin having respective software interfaces indistinguishable from software interfaces of corresponding physical equipment. Optionally, the public endpoints of the energy management system comprise data acquisition endpoints configured to collect data from sensors at the power plant site. Optionally, the model of at least the portion of a power plant site is specified via a site configuration registry that defines a power plant apparatus, electrical connections between the power plant apparatus, and apparatus limits. Optionally, the model of at least a portion of the power plant site is specified via a site configuration registry, the operations further comprising, at least partly in response to determining the model passed testing, deploying the site configuration registry to a physical version of the power plant site. Optionally, software, corresponding to at least a portion of the model, is deployed to operate a physical version of the power plant site during a maintenance window. Optionally, software, corresponding to at least a portion of the model, is deployed to operate a physical version of the power plant site substantially immediately after being made available for deployment. Optionally, the model comprises models of battery energy storage system, a power conditioning system, a meter, and a power breaker.

[0115] Another aspect of the present disclosure relates to an energy management system and methods of using the same. The energy management system is configured to distribute power plant tasks. Energy management system tasks are received at the energy management system. At least a portion of the received tasks are passed over a network from the energy management system to a power plant lead controller in a cluster of nodes, the cluster of nodes comprising electronic devices configured to control power plant equipment, the cluster of nodes comprising the power plant lead controller and a plurality of power plant follower controllers, the cluster of nodes configured to provide a plurality of combinations of redundancy to control power plant equipment. The power plant lead controller delegates, in real time, tasks to the plurality of power plant follower controllers. At least one of the cluster of nodes is configurable to be in hot standby mode.

[0116] Another aspect of the present disclosure relates to a computer-implemented method of distributing tasks at a power plant, the method comprising: receiving energy management system tasks at an energy management system; passing at least a portion of the received tasks over a network from the energy management system to a power plant lead controller in a cluster of nodes,

the cluster of nodes comprising electronic devices configured to control power plant equipment, the cluster of nodes comprising the power plant lead controller and a plurality of power plant follower controllers, the cluster of nodes configured to provide a plurality of combinations of redundancy to control power plant equipment; and delegating, in real time, by the power plant lead controller, tasks to the plurality of power plant follower controllers, wherein at least one of the cluster of nodes, comprising the power plant lead controller and the plurality of power plant follower controllers, is configurable to be in hot standby mode.

[0117] Optionally, at least two power plant follower controllers are located at different physical locations at the power plant. Optionally, at least one node of the cluster of nodes is cloud-based. Optionally, at least one node of the cluster of nodes is located at a second power plant. Optionally, one or more power plant follower controllers comprise microcontrollers configured with real time operating systems. Optionally, the tasks, delegated in real time by the power plant lead controller, comprise resource management tasks. Optionally, the tasks, delegated in real time by the power plant lead controller, comprise taking at least one power plant apparatus out of service. Optionally, method further comprises using cascading commands to propagate commands or actions from one node in the cluster to other nodes in a cascading manner. Optionally, the energy management system comprises a node configured to as operate as a backup to the power plant lead controller. Optionally, the energy management system is configured to manage heterogeneous power sources. Optionally, the method further comprises communicating, by the energy management system, with one or more cluster nodes using an encryption protocol. Optionally, at least one cluster node comprises a server. Optionally, the energy management system comprises containerized code configured to run on virtual machines. Optionally, the energy management system is cloud-based. Optionally, the energy management system is located at the power plant. Optionally, the energy management system is located at a second power plant.

[0118] Another aspect of the present disclosure relates to an energy management system, comprising: a network interface; at least one processing device operable to: receive energy management system tasks; pass at least a portion of the received tasks over a network via the network interface to a power plant lead controller in a cluster of nodes, the cluster of nodes comprising electronic devices configured to control power plant equipment, the cluster of nodes comprising the power plant lead controller and a plurality of power plant follower controllers, the cluster of nodes configured to provide a plurality of combinations of redundancy to control power plant equipment; and cause the power plant lead controller to delegate, in real time, tasks to the plurality of power plant follower controllers, wherein at least one of the cluster of nodes, comprising the power plant lead controller and the plurality of power plant follower controllers, is configurable to be in hot standby mode.

[0119] Optionally, at least two power plant follower controllers are located at different physical locations at the power plant. Optionally, at least one node of the cluster of nodes is cloud-based. Optionally, at least one node of the cluster of nodes is located at a second power plant. Optionally, the tasks, delegated in real time by the power plant lead controller, comprise resource management tasks. Optionally, cascading commands are used to propagate commands or actions from one node in the cluster to other nodes in a cascading manner. Optionally, a node is configured to as operate as a backup to the power plant lead controller. Optionally, the energy management system is configured to manage heterogeneous power sources. Optionally, the energy management system is configured to communicate with one or more cluster nodes using an encryption protocol. Optionally, at least one cluster node comprises a server. Optionally, the system further comprises containerized code configured to run on virtual machines. Optionally, the energy management system is cloud-based. Optionally, the energy management system is located at the power plant. Optionally, the energy management system is located at a second power plant.

[0120] The methods and processes described herein may have fewer or additional steps or states and the steps or states may be performed in a different order. Not all steps or states need to be

reached. The methods and processes described herein may be embodied in, and fully or partially automated via, software code modules executed by one or more general purpose computers. The code modules may be stored in any type of computer-readable medium or other computer storage device. Some or all of the methods may alternatively be embodied in whole or in part in specialized computer hardware. The systems described herein may optionally include displays, user input devices (e.g., touchscreen, keyboard, mouse, voice recognition, etc.), network interfaces, etc.

[0121] The results of the disclosed methods may be stored in any type of computer data repository, such as relational databases and flat file systems that use volatile and/or non-volatile memory (e.g., magnetic disk storage, optical storage, EEPROM and/or solid state RAM).

[0122] The various illustrative logical blocks, modules, routines, and algorithm steps described in connection with the embodiments disclosed herein can be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. The described functionality can be implemented in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the disclosure.

[0123] Moreover, the various illustrative logical blocks and modules described in connection with the embodiments disclosed herein can be implemented or performed by a machine, such as a general purpose processor device, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor device can be a microprocessor, but in the alternative, the processor device can be a controller, microcontroller, or state machine, combinations of the same, or the like. A processor device can include electrical circuitry configured to process computer-executable instructions. In another embodiment, a processor device includes an FPGA or other programmable device that performs logic operations without processing computer-executable instructions. A processor device can also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. Although described herein primarily with respect to digital technology, a processor device may also include primarily analog components. A computing environment can include any type of computer system, including, but not limited to, a computer system based on a microprocessor, a mainframe computer, a digital signal processor, a portable computing device, a device controller, or a computational engine within an appliance, to name a few.

[0124] The elements of a method, process, routine, or algorithm described in connection with the embodiments disclosed herein can be embodied directly in hardware, in a software module executed by a processor device, or in a combination of the two. A software module can reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM, or any other form of a non-transitory computer-readable storage medium. An exemplary storage medium can be coupled to the processor device such that the processor device can read information from, and write information to, the storage medium. In the alternative, the storage medium can be integral to the processor device. The processor device and the storage medium can reside in an ASIC. The ASIC can reside in a user terminal. In the alternative, the processor device and the storage medium can reside as discrete components in a user terminal.

[0125] Conditional language used herein, such as, among others, “can,” “may,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other

embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without other input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list.

[0126] Disjunctive language such as the phrase “at least one of X, Y, Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to present that an item, term, etc., may be either X, Y, or Z, or any combination thereof (e.g., X, Y, and/or Z). Thus, such disjunctive language is not generally intended to, and should not, imply that certain embodiments require at least one of X, at least one of Y, or at least one of Z to each be present.

[0127] While the phrase “click” may be used with respect to a user selecting a control, menu selection, or the like, other user inputs may be used, such as voice commands, text entry, gestures, etc. User inputs may, by way of example, be provided via an interface, such as via text fields, wherein a user enters text, and/or via a menu selection (e.g., a drop down menu, a list or other arrangement via which the user can check via a check box or otherwise make a selection or selections, a group of individually selectable icons, etc.). When the user provides an input or activates a control, a corresponding computing system may perform the corresponding operation. Some or all of the data, inputs and instructions provided by a user may optionally be stored in a system data store (e.g., a database), from which the system may access and retrieve such data, inputs, and instructions. The notifications/alerts and user interfaces described herein may be provided via a Web page, a dedicated or non-dedicated phone application, computer application, a short messaging service message (e.g., SMS, MMS, etc.), instant messaging, email, push notification, audibly, a pop-up interface, and/or otherwise.

[0128] The user terminals described herein may be in the form of a mobile communication device (e.g., a cell phone), laptop, tablet computer, interactive television, game console, media streaming device, head-wearable display, networked watch, etc. The user terminals may optionally include displays, user input devices (e.g., touchscreen, keyboard, mouse, voice recognition, etc.), network interfaces, etc.

[0129] While the above detailed description has shown, described, and pointed out novel features as applied to various embodiments, it can be understood that various omissions, substitutions, and changes in the form and details of the devices or algorithms illustrated can be made without departing from the spirit of the disclosure. As can be recognized, certain embodiments described herein can be embodied within a form that does not provide all of the features and benefits set forth herein, as some features can be used or practiced separately from others. The scope of certain embodiments disclosed herein is indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. (canceled)
2. An energy management system, comprising: at least one processing device; a network interface; and computer readable memory that stores a plurality of hardware interface modules, including a first instantiation of a first hardware interface module and a first instantiation of a second hardware interface module, and instructions that when executed the at least one processing device are

configured to cause the system to perform operations comprising: directly control, via the network interface, a first electronic control device in communication with a first power plant apparatus using the first instantiation of the first hardware interface module stored in the computer readable memory; deploy a second instantiation of the second hardware interface module at a second electronic control device in communication with a second power plant apparatus, wherein the second hardware interface module is configured to control of the second power plant apparatus; detect that the second instantiation of the second hardware interface module at the second electronic control device has failed; and at least partly in response to detecting that the second instantiation of the second hardware interface module at the second electronic control device has failed, execute the first instantiation of the second hardware interface module to control the second power plant apparatus via the network interface and to adjust power and reactive power in real time at least partly in response to changes in grid frequency.

3. The energy management system as defined in claim 2, wherein a plurality of versions of the first electronic control device are located at a first power plant site, wherein the at least one processing device is operable to: select from a plurality of versions of hardware device interface modules stored in memory, the plurality of versions of hardware device interface modules configured for respective versions of the first electronic control device, an appropriate version of the hardware device interface module for a first version of the first electronic control device; and deploy the appropriate version of the hardware device interface module for the first version of the first electronic control device to the first version of the first electronic control device.

4. The energy management system as defined in claim 2, wherein the first power plant apparatus, the first electronic control device, and second electronic control device are located at a first power plant site, wherein the at least one processing device is operable to: enable a third instantiation of the second hardware interface module, hosted by a cloud-based computer system remote from the first power plant site, to enable control of the first power plant apparatus at least partly in response to a failure of the first instantiation of the second hardware interface module and a failure of the second instantiation of the second hardware interface module.

5. The energy management system as defined in claim 2, wherein the second instantiation of the second hardware interface module is maintained in hot standby or on-demand mode.

6. The energy management system as defined in claim 2, wherein the first electronic control device comprises a first server and the second electronic control device comprises a second server.

7. The energy management system as defined in claim 2, wherein the energy management system is configured to manage a plurality of heterogeneous power sources.

8. The energy management system as defined in claim 2, wherein the energy management system is configured to generate instructions and/or recommendations for managing electricity generation, load balancing, the use of backup equipment, and peak demand management.

9. The energy management system as defined in claim 2, wherein the energy management system is configured to mirror data to a remote energy management system.

10. A computer implemented method, comprising: directly controlling, via an energy management system comprising one or more processing devices, a first electronic control device in communication with a first power plant apparatus using a first instantiation of a first hardware interface module stored in computer readable memory; deploying a first instantiation of a second hardware interface module at a second electronic control device in communication with a second power plant apparatus, wherein the second hardware interface module is configured to control of the second power plant apparatus; detecting that the first instantiation of the second hardware interface module at the second electronic control device has failed; and at least partly in response to detecting that the first instantiation of the second hardware interface module at the second electronic control device has failed, execute a second instantiation of the second hardware interface module to control the second power plant apparatus via a network interface.

11. The computer implemented as defined in claim 10, wherein a plurality of versions of the first

electronic control device are located at a first power plant site, the method further comprising: selecting from a plurality of versions of hardware device interface modules stored in memory, the plurality of versions of hardware device interface modules configured for respective versions of the first electronic control device, an appropriate version of the hardware device interface module for a first version of the first electronic control device; and deploying the appropriate version of the hardware device interface module for the first version of the first electronic control device to the first version of the first electronic control device.

12. The computer implemented as defined in claim 10, wherein the first power plant apparatus, the first electronic control device, and second electronic control device are located at a first power plant site, the method further comprising: enabling a third instantiation of the second hardware interface module, hosted by a cloud-based computer system remote from the first power plant site, to enable control of the first power plant apparatus at least partly in response to a failure of the first instantiation of the second hardware interface module and a failure of the second instantiation of the second hardware interface module.

13. The computer implemented as defined in claim 10, the method further comprising maintaining the second instantiation of the second hardware interface module in hot standby or on-demand mode.

14. The computer implemented as defined in claim 10, wherein the first electronic control device comprises a first server and the second electronic control device comprises a second server.

15. The computer implemented as defined in claim 10, the method further comprising managing a plurality of heterogeneous power sources at the same time.

16. The computer implemented as defined in claim 10, the method further comprising generating instructions and/or recommendations for managing electricity generation, load balancing, the use of backup equipment, and peak demand management.

17. The computer implemented as defined in claim 10, the method further comprising mirroring data from the energy management system to a remote energy management system.

18. An offsite energy management system configured to manage a plurality of remote energy generating plants, comprising: at least one processing device; a network interface; and computer readable memory that stores instructions that when executed by the at least one processing device are configured to cause the system to perform operations comprising: communicating with: a first energy management system located at a first energy generating site, and a second energy management system located at a second energy generating site, wherein the first energy management system is configured to directly control a first electronic control device in communication with a first power plant apparatus using a first instantiation of a first hardware interface module; detect that the first energy management system located at the first energy generating site has malfunctioned; and at least partially in response to detecting that the first energy management system located at the first energy generating site has malfunctioned, assume control of the first electronic control device using a second instantiation of the first hardware interface module.

19. The offsite energy management system as defined in claim 18, wherein a plurality of versions of a second electronic control device are located at the first energy generating site, wherein the at least one processing device is operable to: select from a plurality of versions of hardware device interface modules stored in memory, the plurality of versions of hardware device interface modules configured for respective versions of the second electronic control device, an appropriate version of the hardware device interface module for a first version of the second electronic control device; and deploy the appropriate version of the hardware device interface module for the first version of the second electronic control device to the first version of the second electronic control device at the first energy generating site.

20. The offsite energy management system as defined in claim 18, wherein a first instantiation of a second hardware interface module is maintained at a second electronic control device and a second

instantiation of the second hardware interface module is maintained in hot standby or on-demand mode.

21. The offsite energy management system as defined in claim 18, wherein the first electronic control device comprises a first server.

22. The offsite energy management system as defined in claim 18, wherein the offsite energy management system is configured to manage a plurality of heterogeneous power sources.

23. The offsite energy management system as defined in claim 18, wherein the offsite energy management system is configured to generate instructions and/or recommendations for managing electricity generation, load balancing, the use of backup equipment, and peak demand management.

24. The offsite energy management system as defined in claim 18, wherein the offsite energy management system is configured to mirror data to another energy management system.
